WoSM Academy

MAT231-F27 · Calculus II

Implicit and inverse functions

The implicit and inverse function theorems: statement and use

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Last time

- Lagrange: at an extremum of f on $\{g = c\}$ with $\nabla g \neq \mathbf{0}$, $\nabla f = \lambda \nabla g$.
- The multiplier is the rate of change of the optimum: $M'(c) = \lambda$.
- Several constraints: $\nabla f = \sum_j \lambda_j \nabla g_j$, if the ∇g_j are linearly independent.

The proof borrowed one fact: if $\nabla g(\mathbf{a}) \neq \mathbf{0}$, then near $\mathbf{a}$ the set $\{g = c\}$ is a smooth surface. Today: where that fact comes from, and what else it gives.

This lecture

01 Implicit functions

02 The implicit function theorem in general

03 The inverse function theorem

04 Summary and practice

Reading: Marsden & Tromba, *Vector Calculus*, Section 3.5.

This unit is about statements and their use. The proofs are longer than one lecture; we give the idea of each.

01

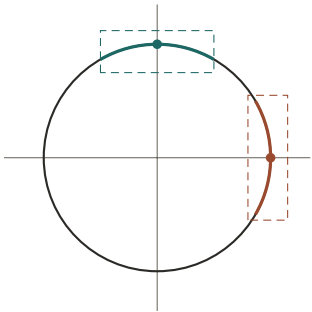
Implicit functions

Curves given by equations

The circle $x^2 + y^2 = 1$ is not the graph of a function $y(x)$. But near most of its points it is:

- near $(0, 1)$: $y = \sqrt{1 - x^2}$;
- near $(0, -1)$: $y = -\sqrt{1 - x^2}$;
- near $(1, 0)$: **no** function $y(x)$: for x just below 1 there are two values of y, for $x > 1$ none. Solve for x instead: $x = \sqrt{1 - y^2}$.

With $F(x, y) = x^2 + y^2$: $F_y = 2y$ vanishes exactly at $(\pm 1, 0)$, where the tangent is vertical.



Near $(0, 1)$ the circle is a graph over the x-axis; near $(1, 0)$ it is not.

Implicit differentiation, revisited

Suppose $F(x, y(x)) = c$ for a differentiable function $y(x)$. The chain rule gives

$$F_x + F_y\, y'(x) = 0, \qquad \text{so} \qquad y'(x) = -\frac{F_x}{F_y} \quad \text{where } F_y \neq 0.$$

Example (The unit circle)

$F = x^2 + y^2$: $y' = -\dfrac{2x}{2y} = -\dfrac{x}{y}$. At $\left(\frac{3}{5}, \frac{4}{5}\right)$ the slope is $-\frac{3}{4}$. At $(\pm 1, 0)$ the formula breaks down — exactly where $y(x)$ does not exist.

Calculus I used this formula assuming that $y(x)$ exists. The theorem says when it does.

The implicit function theorem

Theorem (Implicit function theorem, two variables)

Let F be C^1 near (a, b), with $F(a, b) = c$ and $F_y(a, b) \neq 0$. Then there are open intervals $I \ni a$, $J \ni b$ and a unique function $\varphi\colon I \to J$ such that, inside $I \times J$,

$$F(x, y) = c \quad \Longleftrightarrow \quad y = \varphi(x).$$

φ is C^1, with $\varphi'(x) = -F_x/F_y$ at $(x, \varphi(x))$; if F is C^k, so is φ.

In words: where the level curve is not vertical ($F_y \neq 0$), it is locally the graph of a C^1 function of x.

Why it is true

Say $F_y(a,b) > 0$. By continuity, $F_y > 0$ on a small rectangle around (a,b).

1. For each fixed x, $y \mapsto F(x,y)$ is strictly increasing there. So for each x there is **at most one** y with $F(x,y) = c$.
2. $F(a, b-\delta) < c < F(a, b+\delta)$. By continuity in x, the same inequalities hold for all x close to a.
3. By the intermediate value theorem, for each such x there is **exactly one** $y = \varphi(x)$ between $b-\delta$ and $b+\delta$ with $F(x, \varphi(x)) = c$.

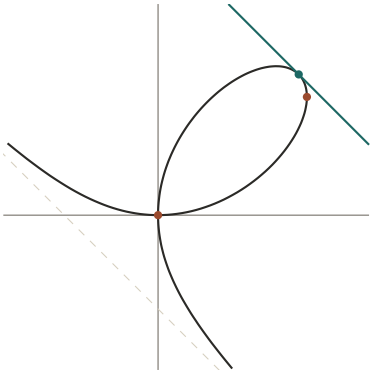
That φ is continuous and differentiable takes more work (Marsden & Tromba, Section 3.5). Once we know φ is differentiable, the formula for φ' is the chain rule.

Example: the folium of Descartes

$F(x, y) = x^3 + y^3 - 6xy = 0$. $F_x = 3x^2 - 6y$, $F_y = 3y^2 - 6x$.

- At $(3, 3)$: $F_x = F_y = 9$, so $y' = -1$. Tangent: $x + y = 6$.
- $F_y = 0$ where $y^2 = 2x$: on the curve, at the origin and at $(2^{5/3}, 2^{4/3}) \approx (3.17, 2.52)$, a vertical tangent.
- At $(0, 0)$, $F_x = F_y = 0$: the curve crosses itself, and is a graph in neither direction.

No formula $y(x)$ is needed to find slopes and tangent lines.



$x^3 + y^3 = 6xy$, its tangent at $(3, 3)$, and the points where $F_y = 0$.

02

The implicit function theorem in general

Surfaces given by an equation

For $F(x, y, z) = c$ with $F_z \neq 0$ at a point, the same theorem lets us solve for $z = \varphi(x, y)$ near it, and

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \qquad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}.$$

Example (The sphere $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$)

$F_z = 2z = 6 \neq 0$, so near $(1, 2, 3)$ the sphere is a graph $z = \varphi(x, y)$, with $z_x = -\frac{x}{z} = -\frac{1}{3}$ and $z_y = -\frac{y}{z} = -\frac{2}{3}$. Tangent plane: $z = 3 - \frac{1}{3}(x - 1) - \frac{2}{3}(y - 2)$, that is $x + 2y + 3z = 14$ — the same as $\nabla F(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}) = 0$ (Lecture 11).

Systems of equations

m equations in $n+m$ unknowns: $F(\mathbf{x},\mathbf{y})=\mathbf{c}$, with $\mathbf{x}\in\mathbb{R}^n$, $\mathbf{y}\in\mathbb{R}^m$ and $F\colon \mathbb{R}^{n+m}\to\mathbb{R}^m$. We want to solve for $\mathbf{y}$ in terms of $\mathbf{x}$.

Theorem (Implicit function theorem)

Let F be C^1 near $(\mathbf{a},\mathbf{b})$, with $F(\mathbf{a},\mathbf{b})=\mathbf{c}$, and suppose the $m\times m$ matrix $D_{\mathbf{y}}F(\mathbf{a},\mathbf{b})=(\partial F_i/\partial y_j)$ is invertible. Then near $(\mathbf{a},\mathbf{b})$ the solutions of $F(\mathbf{x},\mathbf{y})=\mathbf{c}$ are exactly the points $(\mathbf{x},\varphi(\mathbf{x}))$ of the graph of a C^1 function φ, and

$$D\varphi(\mathbf{x}) = -\big(D_{\mathbf{y}}F\big)^{-1} D_{\mathbf{x}}F.$$

The formula: differentiate $F(\mathbf{x},\varphi(\mathbf{x}))=\mathbf{c}$ by the chain rule (Lecture 12): $D_{\mathbf{x}}F + D_{\mathbf{y}}F\,D\varphi = 0$.

The linear model

A linear system $A\mathbf{x} + B\mathbf{y} = \mathbf{c}$ with B an invertible $m \times m$ matrix has the solution $\mathbf{y} = B^{-1}(\mathbf{c} - A\mathbf{x})$.

The idea of the theorem

Near $(\mathbf{a}, \mathbf{b})$ a nonlinear system behaves like its linearisation ($A = D_{\mathbf{x}}F$, $B = D_{\mathbf{y}}F$). If that can be solved for $\mathbf{y}$, so can the nonlinear system — locally.

- m independent equations in $n + m$ unknowns leave an n-dimensional solution set.
- One equation $g = c$ in $\mathbb{R}^n$ with $\nabla g(\mathbf{a}) \neq \mathbf{0}$: some $g_{x_i}(\mathbf{a}) \neq 0$, so we can solve for x_i, and $\{g = c\}$ is a smooth hypersurface near $\mathbf{a}$ (the fact used in Lecture 16). Two equations in $\mathbb{R}^3$ with independent gradients: a smooth curve.

Example: a curve in space

The sphere $x^2 + y^2 + z^2 = 6$ meets the plane $x + y + z = 2$ in a curve through $(1, 2, -1)$. Can we solve for y, z as functions of x?

$$D_{(y,z)}F = \begin{pmatrix} 2y & 2z \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}, \qquad \det = 6 \neq 0. \quad \text{Yes.}$$

Differentiate both equations in x:

$$2x + 2y\,y' + 2z\,z' = 0, \qquad 1 + y' + z' = 0.$$

At $(1, 2, -1)$: $2 + 4y' - 2z' = 0$ and $y' + z' = -1$, so $y' = -\frac{2}{3}$, $z' = -\frac{1}{3}$.

Tangent direction $(1, -\frac{2}{3}, -\frac{1}{3}) \parallel (3, -2, -1)$, perpendicular to both gradients $(2, 4, -2)$ and $(1, 1, 1)$, as it should be.

When the hypothesis fails

Example (The sphere $x^2+y^2+z^2=3$ and the plane $x+y+z=3$)

At $(1,1,1)$: $D_{(y,z)}F = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$, $\det = 0$. The gradients $(2,2,2)$ and $(1,1,1)$ are parallel.

In fact the plane is at distance $3/\sqrt{3} = \sqrt{3}$ from the origin: it **touches** the sphere, and the solution set is the single point $(1,1,1)$ — not a curve.

The theorem is one-directional

If $D_{\mathbf{y}}F$ is singular, the theorem says nothing: the solutions may still be the graph of a function ($y^3 = x$ near the origin, where $F_y = 3y^2 = 0$), or not (above).

03

The inverse function theorem

Inverse functions in one variable

If f is C^1 and $f'(a) \neq 0$, then near a, f is strictly monotone, so it has an inverse, and

$$(f^{-1})'(f(a)) = \frac{1}{f'(a)} \qquad \text{(Calculus I, Unit 11).}$$

If $f'(a) = 0$, anything can happen:

- x^2 near 0: not injective, so no inverse.
- x^3 near 0: injective, but the inverse $\sqrt[3]{y}$ is not differentiable at 0.

In several variables, $f'(a)$ becomes the Jacobian matrix $DF(\mathbf{a})$, and "$\neq 0$" becomes "invertible": $\det DF(\mathbf{a}) \neq 0$.

The inverse function theorem

Theorem

Let $F\colon U \to \mathbb{R}^n$ be C^1 on an open set $U \subseteq \mathbb{R}^n$, and let $DF(\mathbf{a})$ be invertible. Then there are open sets $V \ni \mathbf{a}$ and $W \ni F(\mathbf{a})$ such that $F\colon V \to W$ is a bijection, its inverse $G\colon W \to V$ is C^1, and

$$DG(F(\mathbf{x})) = DF(\mathbf{x})^{-1} \qquad (\mathbf{x} \in V).$$

- The formula is the chain rule applied to $G(F(\mathbf{x})) = \mathbf{x}$: $DG\,DF = I$.
- **Proof idea**: apply the implicit function theorem to $H(\mathbf{x}, \mathbf{y}) = F(\mathbf{x}) - \mathbf{y} = \mathbf{0}$, solving for $\mathbf{x}$ in terms of $\mathbf{y}$. That needs $D_{\mathbf{x}}H = DF(\mathbf{a})$ to be invertible — which is the hypothesis.

Example: polar coordinates

$F(r,\theta) = (r\cos\theta,\ r\sin\theta)$:

$$DF = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \qquad \det DF = r.$$

- $r \neq 0$: F is locally invertible, and $DF^{-1} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\frac{\sin\theta}{r} & \frac{\cos\theta}{r} \end{pmatrix}$. Its first row gives $\frac{\partial r}{\partial x} = \cos\theta = \frac{x}{r}$; the second, $\frac{\partial \theta}{\partial x} = -\frac{\sin\theta}{r} = -\frac{y}{x^2+y^2}$ — without a formula for $\theta(x,y)$.
- $r = 0$: every θ is sent to the origin. No local inverse there.
- **Local, not global**: $F(r,\theta+2\pi) = F(r,\theta)$. Each point has a neighbourhood on which F is injective, but F is not injective on $\{r > 0\}$.

$|\det DF| = r$ is the area factor of polar coordinates (Lecture 20).

Local invertibility everywhere is not enough

Example ($F(x,y) = (e^x \cos y,\ e^x \sin y)$)

$$\det DF = \det \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix} = e^{2x} > 0 \quad \text{everywhere,}$$

so F is locally invertible at every point. But $F(x, y+2\pi) = F(x,y)$: not injective.

- In one variable, $f' \neq 0$ on an interval forces f to be monotone, hence injective. In two variables there is no such argument.
- In complex notation this F is $z \mapsto e^z$, and its local inverses are the branches of the logarithm.

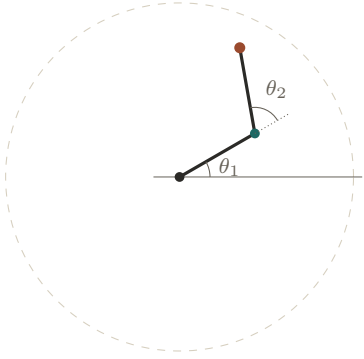
Application: a robot arm

Two links of length 1, joint angles θ_1, θ_2. The hand is at

$$F(\theta_1, \theta_2) = \begin{pmatrix} \cos\theta_1 + \cos(\theta_1 + \theta_2) \\ \sin\theta_1 + \sin(\theta_1 + \theta_2) \end{pmatrix}.$$

A short computation gives $\det DF = \sin\theta_2$.

- $\sin\theta_2 \neq 0$: the angles depend smoothly on the hand position; the hand can move in any direction.
- $\theta_2 = 0$ (stretched) or π (folded): a **singular** configuration. From full stretch the hand cannot move straight outwards.



The hand reaches the disc of radius 2; its edge only with $\theta_2 = 0$.

Application: how a solution moves with a parameter

For each p, the equation $x^3 + x = p$ has exactly one real solution $x(p)$ ($x^3 + x$ is strictly increasing). There is a formula (Cardano's), but we do not need it.

- $F(x, p) = x^3 + x - p$ has $F_x = 3x^2 + 1 > 0$, so $x(p)$ is C^1 (even C^∞), with

$$x'(p) = -\frac{F_p}{F_x} = \frac{1}{3x^2 + 1}.$$

- At $p = 2$: $x = 1$, $x'(2) = \frac{1}{4}$. So $x(2.1) \approx 1.025$ (true value 1.0245).

Remark

The same idea tells how the minimiser of $f(\mathbf{x}, p)$ moves with a parameter p: it solves $\nabla_{\mathbf{x}} f = \mathbf{0}$, and the implicit function theorem applies when the Hessian is invertible.

04

Summary and practice

Summary

- **Implicit function theorem**: $F(\mathbf{x}, \mathbf{y}) = \mathbf{c}$ can be solved locally as $\mathbf{y} = \varphi(\mathbf{x})$ when $D_{\mathbf{y}}F$ is invertible; then $D\varphi = -(D_{\mathbf{y}}F)^{-1} D_{\mathbf{x}}F$. For one equation: $y' = -F_x/F_y$.
- So level sets $\{g = c\}$ with $\nabla g \neq \mathbf{0}$ are smooth hypersurfaces, and m equations with independent gradients define an $(n+m) - m = n$-dimensional set.
- **Inverse function theorem**: $\det DF(\mathbf{a}) \neq 0$ gives a local C^1 inverse, with $DG = DF^{-1}$.
- Both are **local**: polar coordinates and e^z are locally but not globally invertible.
- Uses: tangent lines and planes without formulas; singular configurations; how solutions depend on parameters; change of variables (Lecture 20).

Practice

1. At which points of the ellipse $x^2 - xy + y^2 = 3$ can we *not* solve for y as a function of x?
2. $e^{xy} + x + y = 2$ defines $y(x)$ near $(0, 1)$. Find $y'(0)$.
3. $xz + yz^3 + z = 3$ defines $z(x, y)$ near $(1, 1, 1)$. Find z_x and z_y there.
4. Show that near $(1, 1, 1)$ the system $x^2 + y^2 + z^2 = 3$, $x + y + z = 3$ cannot be solved for (y, z) by the theorem, and find all its solutions.
5. Where is $F(u, v) = (u^2 - v^2,\ 2uv)$ locally invertible? Is it injective on $\mathbb{R}^2 \setminus \{\mathbf{0}\}$?
6. For polar coordinates, find $\partial r/\partial y$ and $\partial\theta/\partial y$ from DF^{-1}.
7. Check that $\det DF = \sin\theta_2$ for the robot arm. Which hand positions are reached in singular configurations?

Answers

1. $F_y = 2y - x = 0$ and the equation give $3y^2 = 3$: the points $(2, 1)$ and $(-2, -1)$.
2. $F_x = ye^{xy} + 1 = 2$, $F_y = xe^{xy} + 1 = 1$: $y'(0) = -2$.
3. $F_x = z = 1$, $F_y = z^3 = 1$, $F_z = x + 3yz^2 + 1 = 5$: $z_x = z_y = -\frac{1}{5}$.
4. $\det \begin{pmatrix} 2y & 2z \\ 1 & 1 \end{pmatrix} = 2y - 2z = 0$ at $(1, 1, 1)$. The only solution is $(1, 1, 1)$: the plane touches the sphere.
5. $\det DF = 4(u^2 + v^2)$: locally invertible except at the origin. Not injective: $F(-u, -v) = F(u, v)$. (It is $z \mapsto z^2$.)
6. $\frac{\partial r}{\partial y} = \sin\theta = \frac{y}{r}$, $\frac{\partial \theta}{\partial y} = \frac{\cos\theta}{r} = \frac{x}{x^2+y^2}$.
7. Expand the 2×2 determinant and use $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$. $\theta_2 = 0$: the circle of radius 2; $\theta_2 = \pi$: the origin.

Next time

Lecture 18 — Double integrals (Unit 18)

- Integrals over rectangles: Riemann sums and volume
- Fubini's theorem: computing by iterated integrals
- General regions, and changing the order of integration

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 5.1 and 5.2.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.